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Abstract

The current epidemiological analysis assesses the seasonal dynamics of hospital admissions and mortality statistics at Kurmitola General Hospital in Dhaka. The data collected over the course of the year reveals some interesting trends in patient inflow and outcomes. For instance, at the beginning of the year, there was a relatively low admission rate, however, as we moved into June, there was a noticeable increase, culminating in more than 470 cases in October. Although the overall mortality rate remained quite low, there was a slight increase in the local mortality rate in the months with a higher admission rate, peaking at 7 in November. This indicates the need for appropriate resource allocation and preparedness far in advance for the time periods with higher admissions. This might also suggest an underlying seasonal pattern driving the patient load. In any case, the analysis in question may serve as a pointer for hospital administrators and healthcare policymakers to improve the delivery of healthcare services.

Introduction

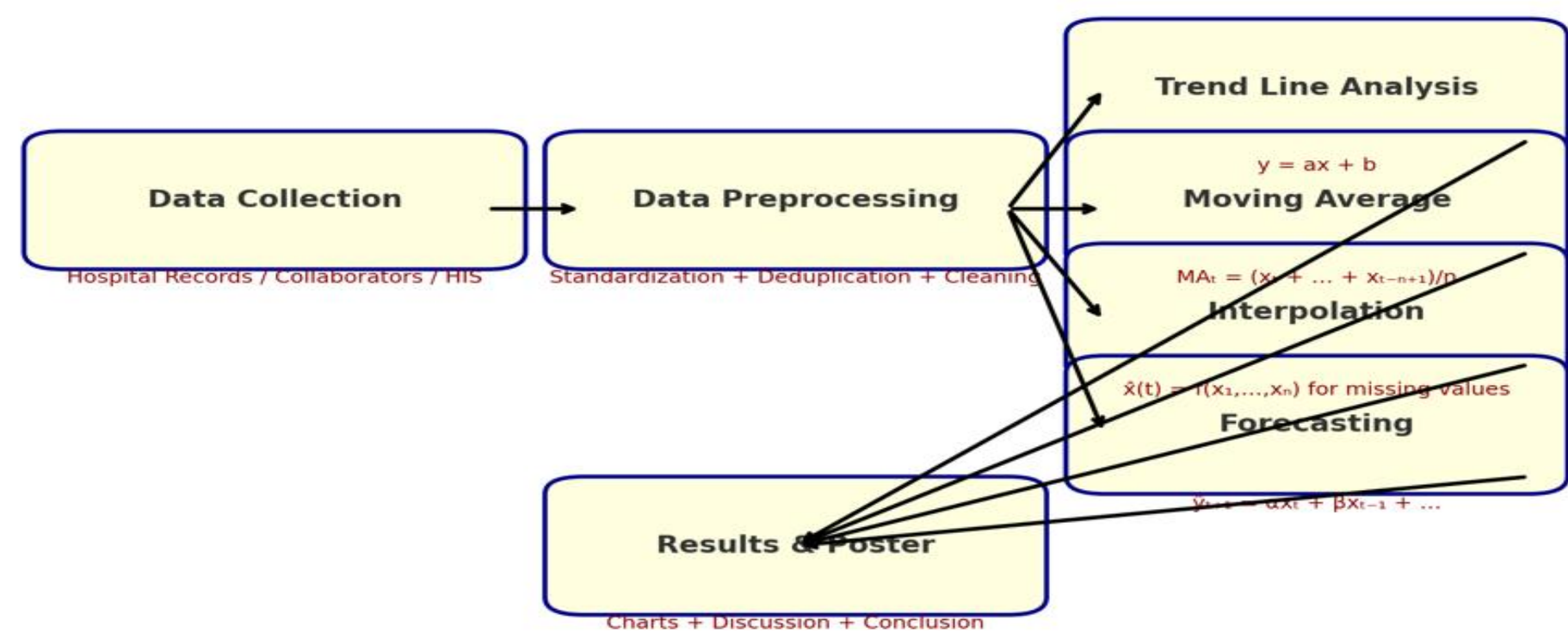
The numbers related to hospital admissions and death rates give us valuable information about the health of a community and how well the local healthcare system is doing. We often see that patient inflow, the burden of diseases, and treatment outcomes can change with the seasons. At Kurmitola General Hospital, the monthly admission records show clear seasonal trends; admissions typically start to rise gradually in the middle of the year and spike towards the end. While overall mortality rates are low, they do tend to increase a bit during the busiest months for admissions. Recognizing these patterns is really important for effectively managing hospital resources and developing policies that aim to improve healthcare services for patients.



Literature Review

Mortality patterns: Mortality is usually low, but slight increases are seen during peak admission periods due to ICU/ward overload and delayed presentation of critical cases.
Operational impact: Seasonal surges require advance planning of beds, staff, oxygen/ventilators, and supplies to avoid service breakdowns.
Data-driven management: Month-wise analysis of admissions and deaths helps identify peak months and supports resource allocation and hospital preparedness.
Local evidence: In Kurmitola General Hospital, admissions peaked in October and mortality in November, showing the importance of seasonal preparedness.
Seasonality in admissions: Many hospitals report seasonal surges in admissions—winter months bring respiratory illnesses, pneumonia, and flu, while monsoon seasons often bring dengue and fever cases in South Asia.

Methodology



Data Collection → Hospital Records / Collaborators / HIS
 $D = \{x_1, x_2, \dots, x_n\}$

Data Cleaning → Standardization + Deduplication
 $D_{\text{clean}} = \text{Standardize}(D) - \text{Duplicates} + \text{Imputation}$

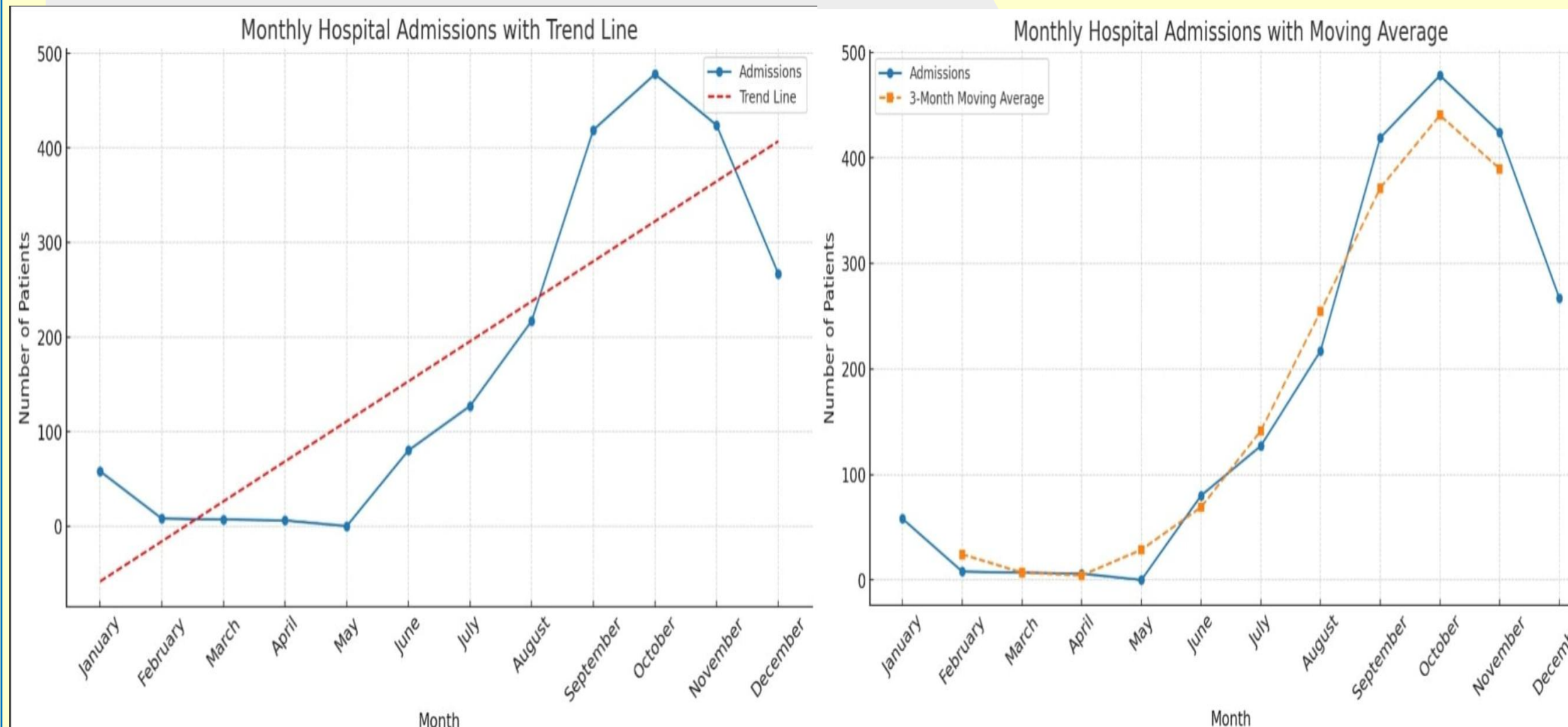
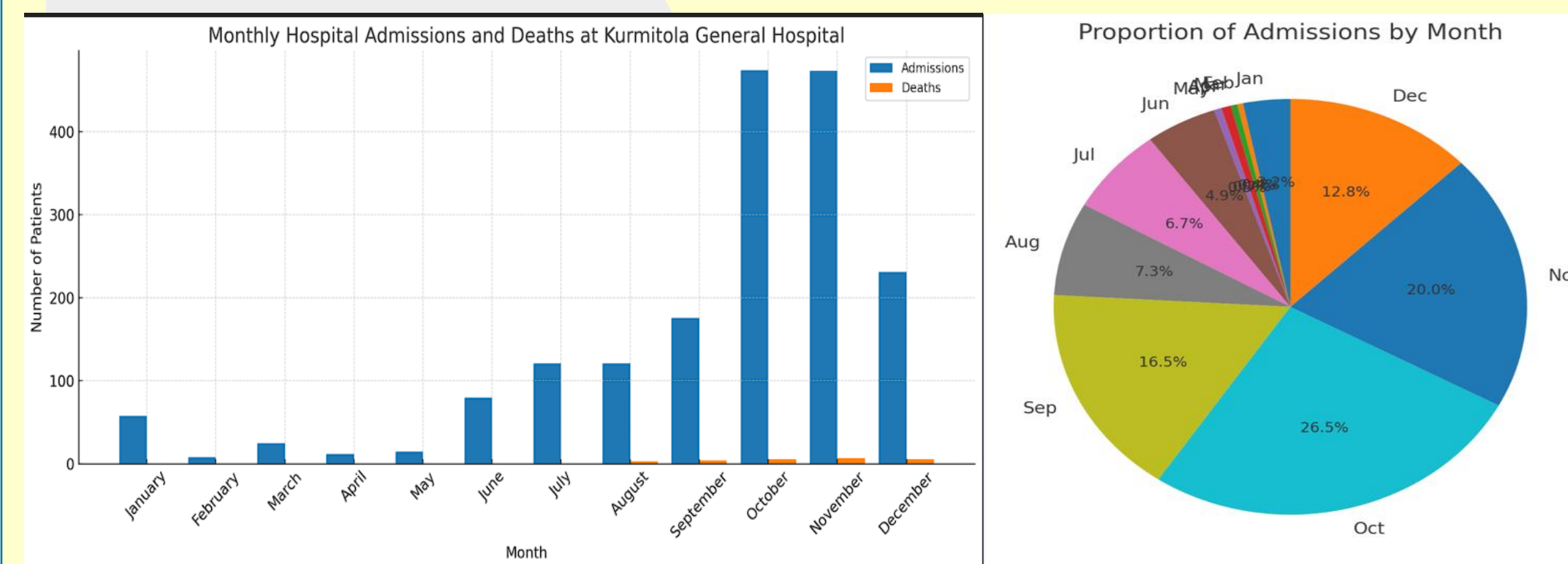
Trend Line Analysis → formula $y = ax + b$

Moving Average → formula $(MA_t = (x_t + \dots + x_{t-n+1})/n)$

Interpolation (Handling Missing Data) → formula $x(t) = (x_i, \dots, x_n)$ for missing values

Forecasting (Predicting Future Values) → formula $y_{\hat{t}+1} = \alpha x_t + \beta x_{t-1} + \dots + \epsilon$

Results



Results: Monthly Hospital Admissions with Trends and Monthly Hospital Admissions with Average Analysis

Monthly Hospital Admissions with Moving Average

This graph uses a **3-Month Moving Average** to smooth out the data.

•The **blue line** shows the raw monthly admissions. You can see sharp fluctuations, with a small dip in March, a steady rise from May to October, and then a drop.

•The **orange dashed line** represents the 3-month moving average. It's a calculated line that averages the current month's data with the two previous months.

•**Purpose:** The moving average reduces the "noise" from the sharp monthly ups and downs. This makes it easier to see the overall seasonal pattern. You can clearly see a major surge in admissions beginning in July, peaking around September–October, and then slowly declining. It provides a less erratic and more stable view of the trend.

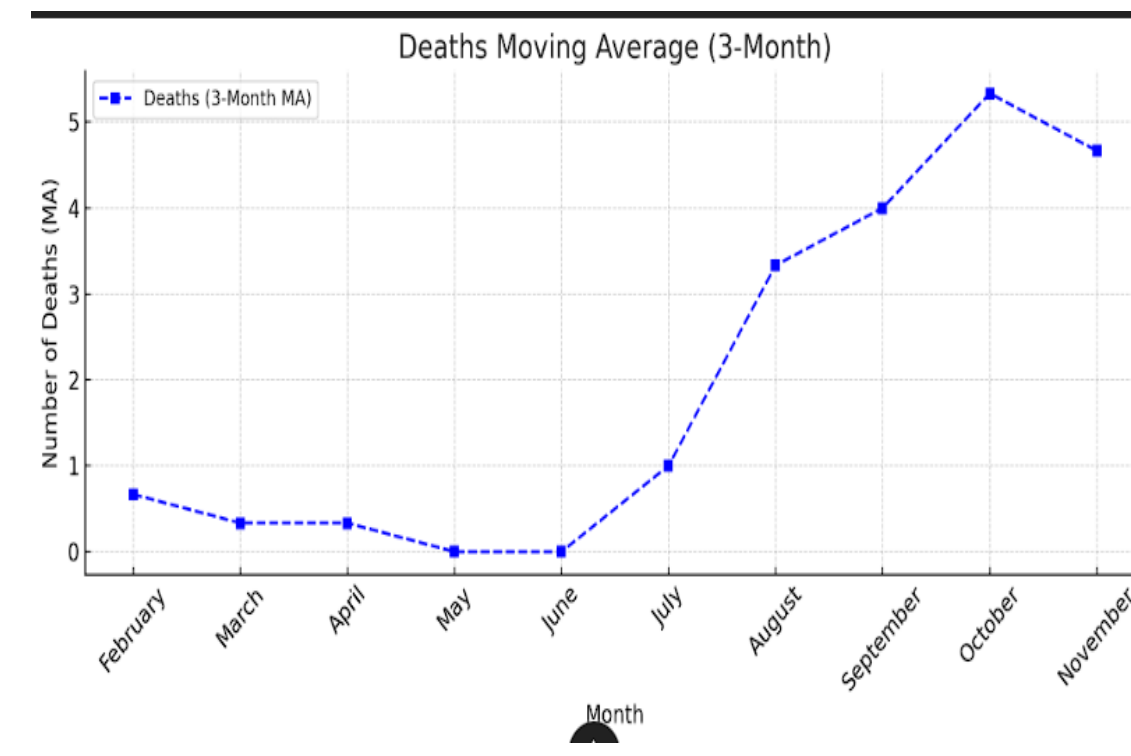
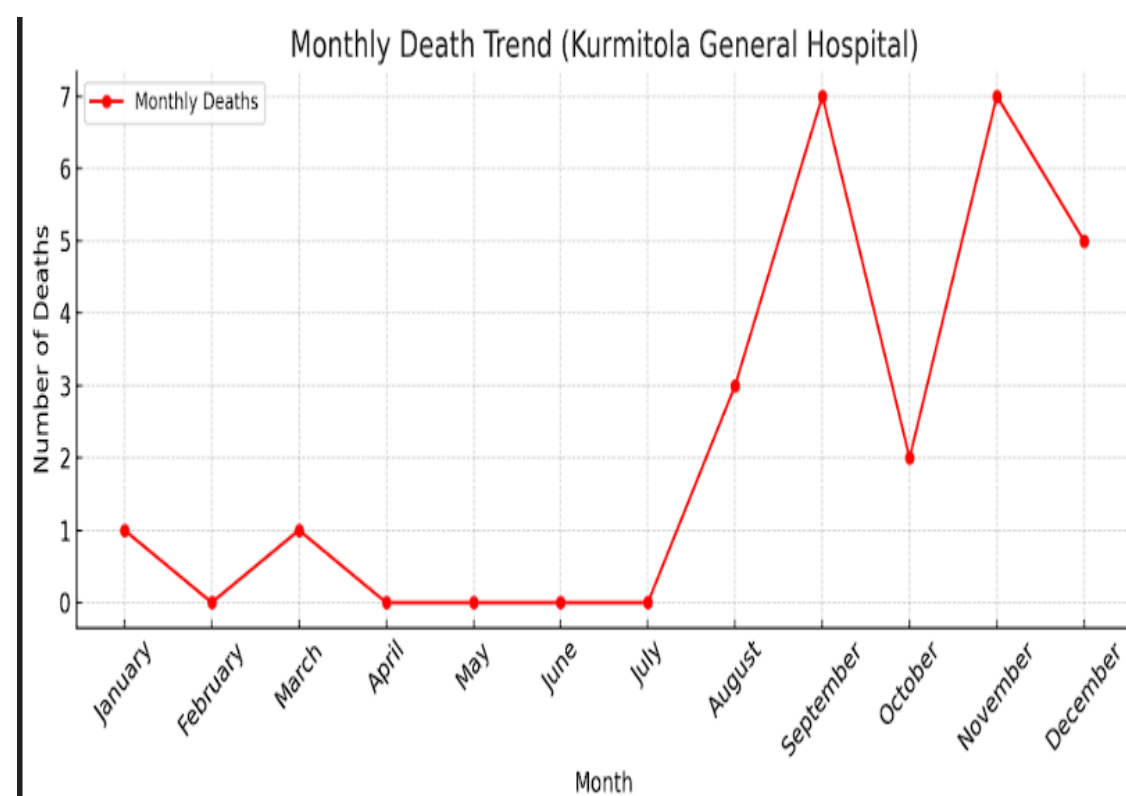
Monthly Hospital Admissions with Trend Line

This graph uses a **Trend Line** to show the long-term direction of the data.

•The **blue line** is the same raw monthly admissions data.

•The **red dashed line** is the trend line. It's a "line of best fit" that calculates the overall upward or downward direction of all the data points.

•**Purpose:** This graph helps you understand the bigger picture. The upward-sloping red line shows that, despite the seasonal peaks and valleys, the general trend for hospital admissions over the entire year is increasing. This is useful for long-range planning and resource management.



Results: Death Trends and 3-Month Moving Average Analysis

Monthly Death Trend

Early Year: Deaths were minimal from February to July (except March), indicating a quiet period.

Mid-to-Late Year Surge: Deaths began to rise in August (3 deaths), peaking in September (7 deaths).

Fluctuation: A dip in October (2 deaths) was followed by another peak in November (7 deaths).

Year-End: December saw a slight decrease to 5 deaths.

Overall: The trend shows a significant increase in deaths towards the end of the year, particularly in September and November.

Deaths 3-Month Moving Average

Initial Stability: The 3-month moving average remained near 0 from February to June, reflecting very low mortality.

Gradual Increase: Starting in July, the moving average began a steady rise (1 death), aligning with the monthly trend.

Peak Average: The moving average reached its highest point in October (5.3 deaths), smoothing out the sharp fluctuations seen in the monthly data.

Slight Decline: A minor decrease to 4.7 deaths was observed in November, still indicating a high level of mortality.

RESULTS

For hospital management and resource planning → Moving Average is better (because it shows seasonal peaks clearly). For long-term performance analysis or forecasting → Trend Line is better (because it shows the overall growth direction).

Discussion

The review of admissions and deaths at Kurmitola General Hospital shows how patient flow changes throughout the year. In the early months, admissions were low, and no deaths were reported. This reflected a quiet period for the hospital. However, starting mid-year, the hospital saw a steady increase in admissions, peaking sharply in October. This sudden rise in cases likely added pressure on healthcare staff and resources. Mortality, though much lower than admissions, followed a similar trend. The highest number of deaths occurred in November, which was one of the busiest months for admissions. This pattern indicates that as patient numbers grow, the likelihood of critical cases and deaths also increases. These findings reveal an important truth: hospitals like Kurmitola must get ready for seasonal surges in patients, especially later in the year. Planning for more beds, staff, and resources during these peak months can significantly reduce mortality and improve patient care. Beyond the statistics, this data serves as a reminder that each "admission" represents a real patient and family in need of timely medical assistance.

Conclusion

This analysis shows that patient admissions at Kurmitola General Hospital increase significantly in the last months of the year. Mortality also peaks during this period. Although the overall death rate stays low, the seasonal rise shows the need for better preparation and resource management in October and November. Most importantly, the data reminds us that behind every number is a patient in need of care. Timely planning can make the difference between life and loss.

Suggestions

- Advanced Forecasting Models – Beyond moving average, methods like ARIMA or Machine Learning can improve accuracy in predicting patient admissions.
- Seasonal Preparedness – Since admissions peak in October–November, hospitals should arrange extra beds, space, and staff during these months.
- Resource Optimization – Data-driven planning can help in efficient allocation of doctors, nurses, and equipment during high-demand periods.
- Policy Support – Government and hospital authorities should implement seasonal surge response plans to reduce overcrowding and mortality risk.

References

- [1] J. A. Freund and R. E. Walpole, Mathematical Statistics, 7th ed. Upper Saddle River, NJ, USA: Prentice Hall, 2000.
- [2] S. Makridakis, S. C. Wheelwright, and R. J. Hyndman, Forecasting: Methods and Applications, 3rd ed. New York, NY, USA: Wiley, 1998.
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